Radiation Maculopathy after Proton Beam Irradiation for Choroidal Melanoma

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Purpose: Radiation maculopathy is a microangiopathy of the retina, which is often observed after irradiation of the eye. To quantitatively determine the frequency of anatomic and functional features of this condition, the authors reviewed a large series of patients with choroidal melanomas who were treated by proton beam irradiation.

Methods: Color photographs and/or fluorescein angiograms of 218 patients with paramacular choroidal melanomas were graded by two independent masked readers to determine the frequency of the various lesions of radiation maculopathy.

Results: Overall, 194 (89%) of the 218 patients in this study developed some degree of radiation maculopathy. The earliest and most common finding was macular edema, which was observed in 87% of patients, overall, by the end of year 3. Microvascular changes (microaneurysms and/or telangiectasia), intraretinal hemorrhages, and capillary nonperfusion were noted in 76%, 70%, and 64% of eyes, respectively, by the end of postirradiation year 3. Visual acuity was retained at 20/200 or better in 90% of eyes 1 year after irradiation and in 67% of eyes 3 years after treatment.

Conclusion: Radiation maculopathy is common after proton beam irradiation of paramacular choroidal melanomas. However, ambulatory vision is preserved in many eyes. *Ophthalmology 1992;99:1278–1285*

Radiation retinopathy is a delayed vaso-occlusive microangiopathy of the retinal vasculature. Its diverse manifestations include intraretinal hemorrhages, cotton-wool spots, microaneurysms, telangiectasia, vascular sheathing, capillary nonperfusion, macular edema, lipid exudation, retinal neovascularization, retinal pigment epithelium changes, and vitreous hemorrhage. Subretinal neovascularization,[1] central retinal vein occlusion,[2] and central retinal artery occlusion[3] are less common findings. In 1933, Stallard[4] first described radiation retinopathy in cases of retinal capillary hemangioma and retinoblastoma treated with radon-seed irradiation. Since then, many investigators have reported radiation retinopathy occurring after local or external beam irradiation.[5-27]

To quantitate the frequency of anatomic and functional features of radiation maculopathy, we studied a large series of patients with paramacular choroidal melanoma who were treated with proton beam irradiation and who were at high risk of radiation maculopathy. Intrinsic tumor factors and damage of the tumor vessels may play a role in post-treatment morbidity and may confuse the picture of radiation injury to normal retinal vessels. To minimize this, we included only eyes with tumors of small and moderate size.

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Patients and Methods

Eyes with choroidal melanomas that were less than 15 mm in diameter, less than 5 mm in height, within 4 disc